Table 17. Execution Time (s) Breakdown Comparison of Baseline and Hierarchical Ensemble Pruning (F-GD)

Team Size	Dataset	CIFAR-10					ImageNet				
	#All Teams	No Pruning	Hierarchical Pruning				No Pruning	Hierarchical Pruning			
		Time (s)	Time (s)	#Teams	Time Reduction (%)	#Teams Reduction (%)	Time (s)	Time (s)	#Teams	Time Reduction (%)	#Teams Reduction (%)
2	45	0.06	0.06	45	**0**	**0**	1.37	1.32	45	**3.65**	**0**
3	120	0.26	0.18	85	**30.77**	**29.17**	9.17	5.82	82	**36.53**	**31.67**
4	210	0.63	0.23	86	**63.49**	**59.05**	28.34	9.78	78	**65.49**	**62.86**
5	252	1.03	0.15	37	**85.44**	**85.32**	52.94	5.32	27	**89.95**	**89.29**

Table 18. Ensemble Execution Cost Reduction by Hierarchical Ensemble Pruning (ImageNet, β = 5%, S_d = 5)

Ensemble Team			Ensemble Acc (%)	Execution Costs		
				#Params (M)	GFLOPs	Inference Time (ms)
Baseline		0123456789	79.82	481.73	149.94	456.00
Top-1 teams for each size S	S=3	245	80.42 (+0.60)	92.64 (**-81%**)	58.50 (**-61%**)	153.69 (**-66%**)
	S=4	2345	80.70 (+0.88)	117.67 (-76%)	67.04 (-55%)	160.93 (65%)
	S=5	12345	**80.77 (+0.95)**	125.65 (-74%)	72.80 (-51%)	208.49 (-54%)

ensemble execution efficiency, i.e., cutting down the storage cost by 74%∼81%, the computing cost by 51%∼61%, and inference latency by 54%∼66%.

6 CONCLUSION

This paper presents our hierarchical ensemble pruning approach powered by the focal diversity metrics. The hierarchical pruning can effectively identify high quality sub-ensembles with a significantly smaller team size and the same or better ensemble accuracy than the entire ensemble team. We made three original contributions. *First,* we optimize ensemble diversity measurements by using focal diversity metrics to accurately capture the failure independence among member networks of a deep ensemble, which closely correlates to ensemble predictive performance and provides effective guidance in ensemble pruning. *Second,* we introduce a hierarchical ensemble pruning algorithm, powered by our focal diversity metrics, which iteratively identifies high quality sub-ensembles and preemptively prunes out the member networks of low diversity. *Third,* we provide a systematic ensemble pruning approach (HQ), which consists of the focal ensemble diversity metric, hierarchical ensemble pruning algorithm and focal diversity consensus voting method. We conducted comprehensive experimental evaluations on four benchmark datasets, CIFAR-10, ImageNet, Cora and MNIST, which demonstrates that our HQ approach can efficiently prune large ensemble teams and obtain high quality sub-ensembles with enhanced ensemble accuracy and significantly reduced execution cost over the entire large ensemble team. Deep ensembles have been widely used in many domain-specific applications, including intelligent medical care [1, 11], intelligent transportation [22, 26], and intelligent manufacturing [12, 14]. One of our future works will focus on applying and optimizing our hierarchical ensemble pruning to enhance the deep ensemble efficiency for these domain-specific applications.

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